Treewidth and the Product of Tree-Independence and Tree-Chromatic Numbers

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Abstract

For a graph G, let $\mathrm{tw}(G)$, $\mathrm{tree}\text{-}\alpha(G)$, and $\mathrm{tree}\text{-}\chi(G)$ denote its treewidth, tree-independence number, and tree-chromatic number, respectively. We prove that every nonempty graph G satisfies

$$\mathrm{tw}(G)+1 \le \mathrm{tree}\text{-}\alpha(G)\,\mathrm{tree}\text{-}\chi(G)\big(1+\lfloor \log_2 \mathrm{tree}\text{-}\alpha(G)\rfloor\big).$$

This improves the quadratic bound of Koutsoutis, Krause, Liu, Redzic, and Ueckerdt [7]. For every pair of integers $n, k \ge 2$, we construct a graph G with $\mathrm{tree}\text{-}\alpha(G) = n$, $\mathrm{tree}\text{-}\chi(G) = k$, and $\mathrm{tw}(G) = \Omega(nk \log_2 n)$. Thus the upper bound is best possible up to a constant factor.

Key Words: Treewidth, tree-independence number, tree-chromatic number

1 Introduction

All graphs in this paper are finite and simple. For a graph G, let $\alpha(G)$, $\omega(G)$, and $\chi(G)$ denote its independence number, clique number, and chromatic number, respectively. For $X \subseteq V(G)$, let $G[X]$ denote the subgraph of G induced by X.

A *tree-decomposition* of G is a pair $\mathcal{T} = (T, \{X_t\}_{t\in V(T)})$, where T is a tree and each $X_t \subseteq V(G)$ is a *bag*, such that:

1. for every edge $uv \in E(G)$, some bag X_t contains both u and v;
2. for every vertex $v \in V(G)$, the set $\{t \in V(T) : v \in X_t\}$ induces a nonempty subtree of T.

The *width* of $\mathcal{T}$ is $\max_{t\in V(T)} |X_t| - 1$. The *treewidth* of G, denoted by $\mathrm{tw}(G)$, is the minimum width over all tree-decompositions of G.

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Treewidth was introduced independently by Halin [4] in 1976 and by Robertson and Seymour [8], who developed it as a part of the theory of graph minors.

The *independence number* of a tree-decomposition $\mathcal{T} = (T, \{X_t\}_{t \in V(T)})$ is $\alpha(\mathcal{T}) := \max_{t \in V(T)} \alpha(G[X_t])$. The minimum of $\alpha(\mathcal{T})$ over all tree-decompositions $\mathcal{T}$ of G is its *tree-independence number*, denoted by $\operatorname{tree}\text{-}\alpha(G)$, which was introduced independently by Yolov [10] and by Dallard, Milanič, and Štorgel [1].

Similarly, the *chromatic number* of $\mathcal{T}$ is $\chi(\mathcal{T}) := \max_{t \in V(T)} \chi(G[X_t])$. The minimum of $\chi(\mathcal{T})$ over all tree-decompositions $\mathcal{T}$ of G is its *tree-chromatic number*, denoted by $\operatorname{tree}\text{-}\chi(G)$, which was introduced by Seymour [9].

Motivated by the elementary inequality $|V(G)| \leq \alpha(G)\chi(G)$, Dallard, Milanič, and Štorgel [1] asked the following question.

Question 1.1 (Dallard, Milanič, and Štorgel [1, Question 8.4])**.** *Is the inequality*

$$\operatorname{tw}(G) + 1 \leq \operatorname{tree}\text{-}\alpha(G) \operatorname{tree}\text{-}\chi(G)$$

valid for every graph G?

Koutsoutis, Krause, Liu, Redzic, and Ueckerdt [7] disproved the inequality in Question 1.1 and established the following quadratic upper bound.

Theorem 1.2 (Koutsoutis et al. [7, Proposition 3])**.** *For every graph G,*

$$\operatorname{tw}(G) + 1 \leq \operatorname{tree}\text{-}\alpha(G)^2 \operatorname{tree}\text{-}\chi(G).$$

Using Seymour's separation lemma [9, Theorem 1.1], we improve Theorem 1.2 as follows.

Theorem 1.3. *Every nonempty graph G satisfies*

$$\operatorname{tw}(G) + 1 \leq \operatorname{tree}\text{-}\alpha(G) \operatorname{tree}\text{-}\chi(G)\big(1 + \lfloor \log_2 \operatorname{tree}\text{-}\alpha(G) \rfloor\big).$$

For the lower bound, Koutsoutis et al. [7] proved the following result.

Theorem 1.4 (Koutsoutis et al. [7, Theorem 2])**.** *Let $f : \mathbb{R} \to \mathbb{R}_{>0}$ and $h : \mathbb{R}_{>0} \to \mathbb{R}_{>0}$ be functions. Suppose that $h(x)/x$ is nondecreasing and $h(x) = o(x \log\log x)$ as $x \to \infty$. Then some graph G satisfies*

$$\operatorname{tw}(G) > h\big(\operatorname{tree}\text{-}\alpha(G)\big) f\big(\operatorname{tree}\text{-}\chi(G)\big).$$

Furthermore, we obtain the following lower bound.

Theorem 1.5. *There is an absolute constant $c > 0$ such that, for all integers $n, k \geq 2$, some graph $G_{n,k}$ satisfies* $\operatorname{tree}\text{-}\alpha(G_{n,k}) = n$, $\operatorname{tree}\text{-}\chi(G_{n,k}) = k$ *and* $\operatorname{tw}(G_{n,k}) + 1 \geq c\, nk \log_2 n$.

Theorem 1.5 shows that the upper bound in Theorem 1.3 is best possible up to a constant factor.

2 Proof of the upper bound

A *separation* of G is a pair (A, B) of subsets of $V(G)$ such that $A \cup B = V(G)$ and there is no edge between $A \setminus B$ and $B \setminus A$. Seymour [9, Theorem 1.1] proved the following lemma.

Lemma 2.1 (Seymour [9, Theorem 1.1])**.** *Every graph G has a separation (A, B) such that $\chi(G[A \cap B]) \le \text{tree-}\chi(G)$ and*

$$\chi(G[A \setminus B]), \chi(G[B \setminus A]) \ge \chi(G) - \text{tree-}\chi(G).$$

Using Lemma 2.1, we prove the following result.

Lemma 2.2. *Every nonempty graph H satisfies*

$$\chi(H) \le \text{tree-}\chi(H)\big(1 + \lfloor \log_2 \alpha(H) \rfloor\big).$$

Proof. We proceed by induction on $\alpha(H)$. If $\alpha(H) = 1$, then the conclusion is immediate. We may therefore assume that $\alpha(H) \ge 2$ and $\chi(H) > \text{tree-}\chi(H)$.

By Lemma 2.1, there is a separation (A, B) of H such that both $H[A \setminus B]$ and $H[B \setminus A]$ have chromatic number at least $\chi(H) - \text{tree-}\chi(H) > 0$ and hence are nonempty. Since there are no edges between these two subgraphs, we have $\alpha(H[A \setminus B]) + \alpha(H[B \setminus A]) \le \alpha(H)$. One of them, say H', therefore satisfies $1 \le \alpha(H') \le \lfloor \alpha(H)/2 \rfloor$. As $\text{tree-}\chi(H') \le \text{tree-}\chi(H)$, using Lemma 2.1 again and the induction hypothesis, we obtain

$$\begin{aligned}\chi(H) &\le \text{tree-}\chi(H) + \chi(H') \le \text{tree-}\chi(H) + \text{tree-}\chi(H')\big(1 + \lfloor \log_2 \alpha(H') \rfloor\big) \\ &\le \text{tree-}\chi(H)\big(2 + \lfloor \log_2 \lfloor \alpha(H)/2 \rfloor \rfloor\big) = \text{tree-}\chi(H)\big(1 + \lfloor \log_2 \alpha(H) \rfloor\big),\end{aligned}$$

as required. □

Proof of Theorem 1.3. Choose a tree-decomposition $\mathcal{T} = (T, \{X_t\}_{t \in V(T)})$ of G with $\alpha(\mathcal{T}) = \text{tree-}\alpha(G)$. Let X_t be a bag of maximum cardinality. Then $\alpha(G[X_t]) \le \text{tree-}\alpha(G)$ and $\text{tree-}\chi(G[X_t]) \le \text{tree-}\chi(G)$. By Lemma 2.2,

$$\begin{aligned}|X_t| &\le \alpha(G[X_t])\chi(G[X_t]) \le \alpha(G[X_t])\,\text{tree-}\chi(G[X_t])\big(1 + \lfloor \log_2 \alpha(G[X_t]) \rfloor\big) \\ &\le \text{tree-}\alpha(G)\,\text{tree-}\chi(G)\big(1 + \lfloor \log_2 \text{tree-}\alpha(G) \rfloor\big).\end{aligned}$$

By the definition of treewidth, we have

$$\text{tw}(G) + 1 \le \text{tree-}\alpha(G)\,\text{tree-}\chi(G)\big(1 + \lfloor \log_2 \text{tree-}\alpha(G) \rfloor\big),$$

as desired. □

3 A matching lower bound

The following construction can raise the treewidth to $|V(H)| - 1$ while preserving the tree-chromatic number of the original graph H and controlling the tree-independence number so that it does not go beyond $\alpha(H)$. We use the following definition and lemma from Koutsoutis, Krause, Liu, Redzic, and Ueckerdt [7, Lemmas 5 and 6]. For a graph H, its 1-*completion* $C(H)$ is obtained by adding, for each pair of nonadjacent vertices $u, v \in V(H)$, a new vertex whose neighborhood is exactly $\{u, v\}$.

Lemma 3.1 (Koutsoutis et al. [7, Lemmas 5 and 6])**.** *Let H be a graph with at least three vertices and at least one edge. Then* tree-$\chi(C(H)) =$ tree-$\chi(H)$, tree-$\alpha(C(H)) \leq \alpha(H)$ *and* $\mathrm{tw}(C(H)) + 1 = |V(H)|$.

For a graph G and a positive integer q, the *lexicographic product* $G[K_q]$, as defined in [5, Section 5.4], is obtained by replacing each vertex $v \in V(G)$ by a clique K_v of order q. For distinct $u, v \in V(G)$, K_u is complete to K_v if and only if $uv \in E(G)$, and anticomplete to K_v otherwise. For a positive integer r, the join $G \vee K_r$ is obtained by adding a clique K_r of order r and making it complete to G; for $r = 0$, this means G itself.

When tree-$\chi(G) = \omega(G)$, we can increase the tree-chromatic number and treewidth together while preserving the tree-independence number. The following lemma extends the treewidth results in Theorem 24 (Section 4.5) and Theorem 22 (Section 4.2) of Gurski and Weishaupt [3] to tree-independence and tree-chromatic numbers.

Lemma 3.2. *Let G be a nonempty graph with* tree-$\chi(G) = \omega(G)$*, and let $q \geq 1$ and $r \geq 0$ be integers. If $H = G[K_q] \vee K_r$, then* tree-$\alpha(H) =$ tree-$\alpha(G)$, tree-$\chi(H) = q$ tree-$\chi(G) + r$ *and* $\mathrm{tw}(H) + 1 = q\big(\mathrm{tw}(G) + 1\big) + r$.

Proof. We first prove the first equality. To show tree-$\alpha(H) \leq$ tree-$\alpha(G)$, choose a tree-decomposition $(T_G, \{U_t\}_{t \in V(T_G)})$ of G with nonempty bags such that $\alpha(G[U_t]) \leq$ tree-$\alpha(G)$ for every $t \in V(T_G)$. Set $T_H := T_G$. For each $t \in V(T_G)$, put $V_t := V(K_r) \cup \bigcup_{v \in U_t} V(K_v)$. Then $(T_H, \{V_t\}_{t \in V(T_H)})$ is a tree-decomposition of H. Since $\alpha(G[U_t]) = \alpha(H[V_t])$ for every $t \in V(T_G)$, we have

$$\text{tree-}\alpha(H) \leq \max_{t \in V(T_H)} \alpha(H[V_t]) = \max_{t \in V(T_G)} \alpha(G[U_t]) = \text{tree-}\alpha(G).$$

Since G is isomorphic to an induced subgraph of H, we have tree-$\alpha(H) \geq$ tree-$\alpha(G)$. Combining this with the preceding inequality gives tree-$\alpha(H) =$ tree-$\alpha(G)$.

We next prove the second equality. To show tree-$\chi(H) \leq q$ tree-$\chi(G) + r$, choose a tree-decomposition $(T'_G, \{W_t\}_{t \in V(T'_G)})$ of G such that $\chi(G[W_t]) \leq$ tree-$\chi(G)$ for every $t \in V(T'_G)$. Set $T'_H := T'_G$. For each $t \in V(T'_G)$, put $X_t := V(K_r) \cup \bigcup_{v \in W_t} V(K_v)$. $(T'_H, \{X_t\}_{t \in V(T'_H)})$ is also a tree-decomposition of H. We have

$$\text{tree-}\chi(H) \leq \max_{t \in V(T'_H)} \chi(H[X_t]) \leq q \max_{t \in V(T'_G)} \chi(G[W_t]) + r = q\,\text{tree-}\chi(G) + r.$$

To show $\operatorname{tree}\text{-}\chi(H) \geq q \operatorname{tree}\text{-}\chi(G) + r$, recall that every clique is contained in a bag of every tree-decomposition (see Diestel [2, Section 12.3, Corollary 12.3.5] for a short proof). Using the hypothesis $\operatorname{tree}\text{-}\chi(G) = \omega(G)$, we obtain

$$\operatorname{tree}\text{-}\chi(H) \geq \omega(H) = q\omega(G) + r = q \operatorname{tree}\text{-}\chi(G) + r.$$

Combining the two inequalities gives $\operatorname{tree}\text{-}\chi(H) = q \operatorname{tree}\text{-}\chi(G) + r$.

Finally, we prove the last equality. To show $\operatorname{tw}(H) + 1 \leq q(\operatorname{tw}(G) + 1) + r$, choose a tree-decomposition $(T''_G, \{Y_t\}_{t \in V(T''_G)})$ of G of width $\operatorname{tw}(G)$. Set $T''_H := T''_G$. For each $t \in V(T''_G)$, put $Z_t := V(K_r) \cup \bigcup_{v \in Y_t} V(K_v)$. Again, $(T''_H, \{Z_t\}_{t \in V(T''_H)})$ is a tree-decomposition of H. We have

$$\operatorname{tw}(H) + 1 \leq \max_{t \in V(T''_H)} |Z_t| = q \max_{t \in V(T''_G)} |Y_t| + r = q\big(\operatorname{tw}(G) + 1\big) + r.$$

To show $\operatorname{tw}(H) + 1 \geq q(\operatorname{tw}(G) + 1) + r$, take a tree-decomposition $(T'''_H, \{A_t\}_{t \in V(T'''_H)})$ of H of width $\operatorname{tw}(H)$. Let T'''_G be the subgraph of T'''_H induced by $\{t \in V(T'''_H) : V(K_r) \subseteq A_t\}$. If $r = 0$, then $T'''_G = T'''_H$. Using the fact that K_r is a clique and the second condition of a tree-decomposition, we conclude that T'''_G is a nonempty subtree of T'''_H. For each $t \in V(T'''_G)$, put $B_t := \{v \in V(G) : V(K_v) \subseteq A_t\}$. For each $v \in V(G)$, the nodes t with $v \in B_t$ form a nonempty intersection of subtrees, since $K_v \vee K_r$ is a clique. For each $uv \in E(G)$, some bag A_t contains the clique $K_u \vee K_v \vee K_r$, so $u, v \in B_t$. Thus $(T'''_G, \{B_t\}_{t \in V(T'''_G)})$ is a tree-decomposition of G.

Choose $t_0 \in V(T'''_G)$ such that $|B_{t_0}|$ is maximum. By the definition of treewidth,

$$\operatorname{tw}(G) + 1 \leq \max_{t \in V(T'''_G)} |B_t| = |B_{t_0}|.$$

The bag A_{t_0} contains the pairwise disjoint vertex sets $V(K_v)$ for all $v \in B_{t_0}$ and $V(K_r)$. Since each K_v has order q, we have $q|B_{t_0}| + r \leq |A_{t_0}|$. The chosen tree-decomposition of H has width $\operatorname{tw}(H)$, so $|A_{t_0}| \leq \operatorname{tw}(H) + 1$. Consequently,

$$q\big(\operatorname{tw}(G) + 1\big) + r \leq q|B_{t_0}| + r \leq |A_{t_0}| \leq \operatorname{tw}(H) + 1.$$

Combining the two inequalities gives $\operatorname{tw}(H) + 1 = q(\operatorname{tw}(G) + 1) + r$. □

The following construction is inspired by the shift graphs in Huynh, Reed, Wood, and Yepremyan [6, Section 2].

For convenience, we restate Theorem 1.5 here.

Theorem 1.5. *There is an absolute constant $c > 0$ such that, for all integers $n, k \geq 2$, some graph $G_{n,k}$ satisfies $\operatorname{tree}\text{-}\alpha(G_{n,k}) = n$, $\operatorname{tree}\text{-}\chi(G_{n,k}) = k$ and $\operatorname{tw}(G_{n,k}) + 1 \geq c\,nk \log_2 n$.*

Proof. Fix an integer $n \geq 3$. Let

$$\mathcal{I}_n := \big\{[x, x + 3^i] : i, x \in \mathbb{Z},\ 0 \leq i \leq \lfloor \log_3 n \rfloor,\ 0 \leq x \leq n - 3^i\big\}.$$

Define H_n to be the graph with vertex set $\mathcal{I}_n$ in which two distinct vertices are adjacent if and only if they can be written as $[x, y]$ and $[z, w]$ such that

$$x < z \leq y < w \qquad \text{and} \qquad y - z \in \{0, 1, 3, 9, \ldots\}. \tag{1}$$

Claim 1.5.1. $\alpha(H_n) \leq n$.

Proof. Let $\mathcal{S}$ be an independent set of H_n. Form a graph F on $\{0, \ldots, n\}$ by including the edge xy for every $[x, y] \in \mathcal{S}$, and orient each edge from its smaller endpoint to its larger endpoint. There is no directed path $x \to y \to z$ of length two, since $[x, y]$ and $[y, z]$ would be adjacent in H_n. Hence every nonisolated vertex of F is either a source or a sink.

Suppose that C is a cycle in F. Choose the smallest sink u on C, and let $\ell < r$ be its two neighbors on C. Let v be the other neighbor of r on C. The vertex r is a source, so v is a sink, and the choice of u gives $\ell < r < u < v$. Since $[r, u]$ is a vertex of H_n, the construction of H_n implies that $u - r$ is a power of three. By (1), the vertices $[\ell, u]$ and $[r, v]$ are adjacent, a contradiction. Thus F is a forest, and $|\mathcal{S}| = |E(F)| \leq n$. □

Claim 1.5.2. *H_n has a tree-decomposition of chromatic number at most two.*

Proof. Let P be the path with vertices $0, \ldots, n$ in this order. For each $t \in V(P)$, put $X_t := \{[x, y] \in V(H_n) : x \leq t \leq y\}$. Clearly, $(P, \{X_t\}_{t \in V(P)})$ is a tree-decomposition of H_n.

We show that $\chi(H_n[X_t]) \leq 2$ for every $t \in V(P)$. Fix $t \in V(P)$. Give a vertex $[x, y] \in X_t$ color 1 if $t < (x + y)/2$, and color 2 if $t > (x + y)/2$. Since $y - x$ is odd and t is an integer, $(x + y)/2 \neq t$, so every vertex receives a color.

Let $[x, y]$ and $[z, w]$ be adjacent vertices in X_t, labelled so that $x < z \leq y < w$. Since both vertices belong to X_t, we have $z \leq t \leq y$. If $z = t = y$, then $(x + y)/2 < t < (z + w)/2$. Otherwise, $z < y$. By (1) and the definition of H_n, $y - x$, $y - z$, and $w - z$ are powers of three. $y - z$ is smaller than both $y - x$ and $w - z$, so each interval has length at least three times $y - z$, and hence $(x + y)/2 < t < (z + w)/2$. Thus $[x, y]$ receives color 2 and $[z, w]$ receives color 1. Therefore, this coloring is proper. Thus $\chi(H_n[X_t]) \leq 2$ for every $t \in V(P)$, as required. □

Counting intervals of lengths 3^i with $0 \leq i < \lfloor \log_3 n \rfloor$ gives

$$|V(H_n)| \geq \sum_{i=0}^{\lfloor \log_3 n \rfloor - 1} (n + 1 - 3^i) \geq \sum_{i=0}^{\lfloor \log_3 n \rfloor - 1} \frac{2n}{3} = \frac{2n}{3} \lfloor \log_3 n \rfloor \geq \frac{n \log_2 n}{3 \log_2 3},$$

since $i < \lfloor \log_3 n \rfloor$ implies $3^i \leq n/3$, and $\lfloor \log_3 n \rfloor \geq \frac{1}{2} \log_3 n$ for $n \geq 3$.

The vertices $[0, 1]$ and $[1, 2]$ are adjacent, so Claim 1.5.2 gives tree-$\chi(H_n) = 2$. Applying Lemma 3.1 together with Claim 1.5.1 yields tree-$\chi(C(H_n)) = 2$, tree-$\alpha(C(H_n)) \leq n$ and $\mathrm{tw}(C(H_n)) + 1 = |V(H_n)|$.

For $n \geq 3$, let B_n be the disjoint union of $C(H_n)$ and $K_{n,n}$. Since tree-$\chi(K_{n,n}) = 2$, $\mathrm{tw}(K_{n,n}) = n$ and tree-$\alpha(K_{n,n}) = n$, taking the maximum over components gives tree-$\alpha(B_n) = n$, tree-$\chi(B_n) = 2$ and $\mathrm{tw}(B_n) + 1 = \max\{|V(H_n)|, n+1\}$. There are n intervals of length one and $n-2$ of length three, so $|V(H_n)| \geq 2n - 2 \geq n + 1$. Hence $\mathrm{tw}(B_n) + 1 = |V(H_n)|$. Also, B_n has an edge and $\omega(B_n) \leq$ tree-$\chi(B_n) = 2$, so $\omega(B_n) = 2$.

For $n = 2$, set $B_2 := K_{2,2}$, so tree-$\alpha(B_2) =$ tree-$\chi(B_2) = 2$ and $\mathrm{tw}(B_2)+1 = 3$. Together with the lower bound on $|V(H_n)|$, this gives, for every integer $n \geq 2$, tree-$\alpha(B_n) = n$, tree-$\chi(B_n) = \omega(B_n) = 2$ and $\mathrm{tw}(B_n) + 1 \geq n \log_2 n/(3 \log_2 3)$.

Now fix integers $n \geq 2$ and $k \geq 2$, and put $q := \lfloor k/2 \rfloor$ and $r := k - 2q \in \{0, 1\}$. Define $G_{n,k} := B_n[K_q] \vee K_r$. Since tree-$\chi(B_n) = \omega(B_n) = 2$, Lemma 3.2 gives tree-$\alpha(G_{n,k}) = n$ and tree-$\chi(G_{n,k}) = 2q + r = k$. Since $q \geq k/3$ and $r \geq 0$, the treewidth formula in the same lemma gives

$$\mathrm{tw}(G_{n,k}) + 1 = q\big(\mathrm{tw}(B_n) + 1\big) + r \geq \frac{k}{3} \cdot \frac{n \log_2 n}{3 \log_2 3} = \frac{nk \log_2 n}{9 \log_2 3}.$$

This proves the theorem with $c = 1/(9 \log_2 3)$. □

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References

[1] C. Dallard, M. Milanič, and K. Štorgel, *Treewidth versus clique number. II. Tree-independence number*, J. Combin. Theory Ser. B **164** (2024), 404–442, doi:10.1016/j.jctb.2023.10.006.

[2] R. Diestel, *Graph Theory*, 5th ed., Graduate Texts in Mathematics **173**, Springer, Berlin, 2017, doi:10.1007/978-3-662-53622-3.

[3] F. Gurski and R. Weishaupt, *The behavior of tree-width and path-width under graph operations and graph transformations*, Algorithms **18**(7) (2025), 386, doi:10.3390/a18070386.

[4] R. Halin, *S-functions for graphs*, J. Geom. **8** (1976), 171–186, doi:10.1007/BF01917434.

[5] R. Hammack, W. Imrich, and S. Klavžar, *Handbook of Product Graphs*, 2nd ed., CRC Press, Boca Raton, FL, 2011, doi:10.1201/b10959.

[6] T. Huynh, B. Reed, D. R. Wood, and L. Yepremyan, *Notes on tree- and path-chromatic number*, in *2019–20 MATRIX Annals*, MATRIX Book Ser. **4**, Springer, Cham, 2021, 489–498, doi:10.1007/978-3-030-62497-2_30.

[7] A. Koutsoutis, K. Krause, C.-H. Liu, M. Redzic, and T. Ueckerdt, *On the relation between treewidth, tree-independence number, and tree-chromatic number of graphs*, in *52nd International Workshop on Graph-Theoretic Concepts in Computer Science (WG 2026)*, Leibniz Int. Proc. Inform. **376** (2026), 31:1–31:9, doi:10.4230/LIPIcs.WG.2026.31.

[8] N. Robertson and P. D. Seymour, *Graph minors. II. Algorithmic aspects of tree-width*, J. Algorithms **7** (1986), 309–322, doi:10.1016/0196-6774(86)90023-4.

[9] P. Seymour, *Tree-chromatic number*, J. Combin. Theory Ser. B **116** (2016), 229–237, doi:10.1016/j.jctb.2015.08.002.

[10] N. Yolov, *Minor-matching hypertree width*, in *Proceedings of the Twenty-Ninth Annual ACM-SIAM Symposium on Discrete Algorithms (SODA 2018)*, SIAM, Philadelphia, 2018, 219–233, doi:10.1137/1.9781611975031.16.